Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Linear Inequalities with Two Variables

Name: ________________
Date: ____________

Advanced (Carolina Reaper)

Q1. What is the graph of the inequality $y > 2x - 3$?
(A) A line with a slope of 2 and y-intercept of -3
(B) A line with a slope of 2 and y-intercept of 3
(C) A shaded region above the line $y = 2x - 3$
(D) A shaded region below the line $y = 2x - 3$
(E) A single point (0, -3)

Q2. Which inequality has a solid boundary line?
(A) $ygeq2x - 3$
(B) $yleq2x - 3$
(C) $y > 2x - 3$
(D) $y < 2x - 3$
(E) $y = 2x - 3$

Q3. The inequality $x + yleq4$ has a boundary line with a slope of
(A) -1
(B) 1
(C) 0
(D) 4
(E) -4

Q4. Which region is shaded for the inequality $y < -2x + 1$?
(A) Above the line $y = -2x + 1$
(B) Below the line $y = -2x + 1$
(C) To the left of the line $y = -2x + 1$
(D) To the right of the line $y = -2x + 1$
(E) Between the lines $y = -2x + 1$ and $y = 2x - 1$

Q5. The inequality $x > 0$ represents a
(A) Vertical line with equation $x = 0$
(B) Horizontal line with equation $y = 0$
(C) Shaded region to the right of the y-axis
(D) Shaded region to the left of the y-axis
(E) Single point (0, 0)

Q6. Which of the following is equivalent to the inequality $ygeq - x + 2$?
(A) $-yleqx - 2$
(B) $yleqx - 2$
(C) $ygeqx + 2$
(D) $-ygeqx + 2$
(E) $y = x + 2$

Q7. The solution to the inequality $x+2yleq6$ can be represented by a
(A) Line with equation $x + 2y = 6$
(B) Shaded region above the line $x + 2y = 6$
(C) Shaded region below the line $x + 2y = 6$
(D) Single point (0, 3)
(E) Line segment with endpoints (0, 0) and (6, 0)

Q8. The inequality $y < 0$ represents a
(A) Shaded region above the x-axis
(B) Shaded region below the x-axis
(C) Vertical line with equation $x = 0$
(D) Horizontal line with equation $y = 0$
(E) Single point (0, 0)

Open Ended Questions (FRQ)

Q9. Graph the solution to the inequality $y > 2x - 3$ and identify the boundary line.

Q10. Find the solution to the system of inequalities $x+yleq4$ and $x-ygeq-2$. Graph the solution region and identify any vertices.

Chef's Tips

- Emphasize the importance of shading the correct region.
- Remind students to use a solid line for *geq* and *leq* inequalities.
- Encourage students to check their work by graphing the boundary line and testing points in each region.